

MCALLEN, TX, USA

WMO: 722506

Lat: 26.176N

Lon: 98.240W

Elev: 31

StdP: 100.96

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12959

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	5.5	-7.4	2.0	13.6	-4.3	2.6	13.5	12.1	23.0	11.2	22.8	4.0	340	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.0	38.5	24.7	37.6	24.9	36.7	24.8	26.9	32.9	26.6	32.4	26.3	31.7	6.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.3	28.1	25.4	20.7	28.1	25.1	20.3	28.0	84.6	33.0	83.0	31.9	81.7	31.6	33.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.2	10.2	9.2	DB	0.8	40.4	2.0	1.6	-0.6	41.5	-1.8	42.4	-2.9	43.3	-4.4	44.5
			WB	-1.0	28.1	1.6	1.3	-2.2	29.0	-3.2	29.7	-4.1	30.4	-5.3	31.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.6	16.6	19.0	21.9	25.1	28.1	30.3	30.8	31.1	28.8	25.4	20.9	17.2	
	DBStd	6.35	5.17	5.41	4.47	3.46	2.50	1.59	1.66	1.69	2.23	3.67	4.64	5.17	
	HDD10.0	22	10	4	1	0	0	0	0	0	0	0	1	6	
	HDD18.3	287	93	54	21	3	0	0	0	0	0	3	28	85	
	CDD10.0	5370	216	257	368	454	562	608	646	656	564	479	329	229	
	CDD18.3	2593	41	73	130	207	304	358	388	397	314	224	106	50	
	CDH23.3	28534	266	512	951	1848	3348	4511	5029	5193	3555	2183	808	329	
CDH26.7	13891	57	168	337	763	1579	2404	2764	2886	1693	912	251	77		
Wind	WSAvg	4.7	4.3	4.8	5.2	5.5	5.6	5.3	5.4	4.7	3.6	4.0	4.0	4.0	
Precipitation	PrecAvg	584	32	30	19	35	74	71	40	60	110	62	27	29	
	PrecMax	907	118	133	96	150	201	266	212	205	277	190	106	99	
	PrecMin	219	0	0	0	0	0	0	0	0	8	1	0	0	
	PrecStd	166	32	27	25	39	57	67	50	52	71	52	28	28	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.5	33.6	34.9	36.3	37.4	38.7	39.6	39.7	38.6	35.8	33.2	30.9	
		MCWB	20.1	20.5	20.9	23.2	25.5	25.3	24.6	24.9	24.0	24.6	23.6	21.6	
	2%	DB	28.2	30.8	31.9	34.1	35.7	37.3	38.2	38.6	36.7	34.1	31.1	28.8	
		MCWB	19.8	21.0	22.1	23.2	25.2	25.3	24.8	24.9	24.6	24.1	22.3	20.9	
	5%	DB	26.5	28.7	30.2	32.6	34.6	36.4	37.3	37.6	35.2	32.9	29.5	27.0	
		MCWB	19.0	20.6	21.3	23.1	24.8	25.1	24.9	24.9	24.6	23.8	21.6	20.3	
	10%	DB	24.3	26.6	28.4	31.1	33.4	35.4	36.2	36.6	33.8	31.5	27.9	24.9	
		MCWB	18.3	19.5	20.5	22.4	24.4	24.9	24.8	24.8	24.6	23.5	21.0	19.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.4	23.5	24.1	26.2	27.4	27.4	27.0	26.9	27.1	26.9	25.3	23.5
			MCDB	26.8	29.3	30.5	32.8	34.9	34.9	33.3	33.0	31.6	30.9	29.7	27.7
		2%	WB	21.4	22.6	23.4	25.2	26.5	26.9	26.6	26.6	26.6	26.2	24.2	22.7
			MCDB	25.5	27.4	28.8	31.2	33.3	33.3	32.5	32.5	31.2	30.1	28.0	26.4
5%		WB	20.7	21.8	22.7	24.4	25.8	26.5	26.2	26.2	26.2	25.7	23.4	21.8	
		MCDB	24.3	26.1	27.4	29.7	31.7	32.2	31.9	32.0	30.8	29.6	27.1	25.2	
10%		WB	19.8	21.1	22.0	23.8	25.4	26.1	25.8	25.8	25.8	25.1	22.6	20.9	
		MCDB	23.0	24.8	26.1	28.6	30.7	31.5	31.4	31.5	30.4	29.0	26.2	24.1	
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.1	10.8	10.8	10.2	10.6	10.8	11.0	10.0	10.7	10.8	10.6	
		MCDBR	12.8	12.7	11.9	11.8	11.2	11.6	12.0	12.1	11.6	11.2	11.6	12.0	
		MCWBR	6.2	5.0	4.3	4.1	3.0	2.3	2.4	2.4	2.6	3.5	4.5	6.0	
	5% WB	MCDBR	10.6	10.6	10.1	10.2	9.8	10.2	10.3	10.7	9.4	9.3	9.8	10.0	
		MCWBR	5.6	4.9	4.2	3.9	3.1	2.5	2.4	2.5	2.7	3.6	4.8	5.7	
Clear-Sky Solar Irradiance	taub	0.344	0.363	0.390	0.412	0.414	0.407	0.419	0.411	0.416	0.376	0.366	0.344		
	taud	2.494	2.430	2.358	2.310	2.343	2.415	2.384	2.418	2.373	2.479	2.455	2.500		
	Ebn at Noon	908	914	904	889	881	880	869	878	868	890	875	893		
	Edh at Noon	95	109	123	132	128	118	122	117	120	102	98	91		
All-Sky Solar Radiation	RadAvg	3.19	3.88	4.73	5.56	6.02	6.73	6.68	6.41	5.31	4.61	3.58	2.89		
	RadStd	0.36	0.59	0.56	0.52	0.42	0.25	0.47	0.30	0.41	0.46	0.40	0.42		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d) +0.47	(e) N/A	(f) N/A	(g) +0.62	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +198	(m) +164
Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95

Nomenclature: See separate page